Geometric Sequences and Series

ANSWER KEY

Geometric sequences

1 Consider the sequence $3, 6, 12, 24, \ldots$

a State the common ratio. $r = 2$

b Write a formula for the nth term. $a_n = 3 \cdot 2^{n-1}$

c Find a_{10}. $a_{10} = 1536$

2 In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.

$r^3 = \frac{48}{6} = 8$, so $r = 2$ and $a_1 = 3$.

Geometric series

3 Find the sum of the first 8 terms of $5 + 10 + 20 + \cdots$

$S_8 = 5\frac{2^8 - 1}{2 - 1} = 5(255) = 1275$.

4 Evaluate each infinite sum, or state that it diverges:

a $18 + 6 + 2 + \frac{2}{3} + \cdots$ $r = \frac{1}{3}$: $S = \frac{18}{1 - 1/3} = 27$.

b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$ $r = -\frac{1}{2}$: $S = \frac{1}{1 + 1/2} = \frac{2}{3}$.

c $2 + 3 + 4.5 + \cdots$ $r = 1.5 > 1$: diverges.

5 Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.

$0.7 + 0.07 + \cdots$ with $a = \frac{7}{10}$, $r = \frac{1}{10}$: $S = \frac{7/10}{9/10} = \frac{7}{9}$.

6 A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.

Total $= 8 + 2\left(8 \cdot \frac{3}{4} + 8 \cdot \left(\frac{3}{4}\right)^2 + \cdots\right) = 8 + 2\frac{6}{1 - 3/4} = 8 + 48 = 56$ m.

Geometric means and mixed sequences

7 Insert two positive geometric means between 4 and 108 (i.e., find b_1, b_2 so that $4, b_1, b_2, 108$ is geometric).

4 terms total: $r^3 = \frac{108}{4} = 27$, so $r = 3$. Means: $12, 36$.

8 Determine whether each sequence is arithmetic, geometric, or neither:

a $2, 6, 18, 54, \ldots$ Geometric ($r = 3$)

b $5, 8, 11, 14, \ldots$ Arithmetic ($d = 3$)

c $1, 4, 9, 16, \ldots$ Neither (differences $3, 5, 7$ are not constant, and ratios are not constant)

Applications

9 A car worth 80,000 riyals depreciates by 15% per year.

a Write a formula for its value after n years. $V_n = 80000(0.85)^n$.

b Find its value after 5 years, to the nearest riyal. $V_5 = 80000(0.85)^5 \approx 80000(0.4437) \approx 35{,}497$ riyals.

10 A chain letter starts with 1 person sending it to 5 people, each of whom sends it to 5 more, and so on. Find the total number of people who have received the letter after 6 rounds (including the original sender's round).

This is a geometric series with $a = 1$, $r = 5$, $n = 6$: $S_6 = \frac{1(5^6 - 1)}{5 - 1} = \frac{15624}{4} = 3906$.

Visualizing geometric growth

11 The first 6 terms of $a_n = 2(1.5)^{n-1}$ are plotted below. Contrast the shape of this graph with the straight-line shape of an arithmetic sequence's graph.

Unlike an arithmetic sequence's straight-line pattern, the geometric sequence's points curve upward with increasing steepness — each term is a constant MULTIPLE of the previous one rather than a constant ADDITION, producing exponential-style curvature rather than a line.

Determining convergence before summing

12 For each infinite geometric series, determine whether it converges WITHOUT computing the sum, by checking $|r| < 1$: $\sum_{n=1}^{\infty} 5(0.9)^{n-1}$, $\sum_{n=1}^{\infty} 3(1.2)^{n-1}$, $\sum_{n=1}^{\infty} 7(-0.4)^{n-1}$.

First: $r = 0.9$, $|r| < 1$: converges. Second: $r = 1.2$, $|r| > 1$: diverges. Third: $r = -0.4$, $|r| < 1$: converges.

Finding the number of terms from a partial sum

13 For the geometric series $2 + 6 + 18 + \cdots$, find the number of terms n needed for the sum to first exceed 2000.

$S_n = \frac{2(3^n - 1)}{3 - 1} = 3^n - 1 > 2000 \Rightarrow 3^n > 2001$. Testing: $3^6 = 729$, $3^7 = 2187$. So $n = 7$ terms are needed (giving $S_7 = 3^7 - 1 = 2186 > 2000$).

A mixed application: population with geometric decline

14 A population of an endangered species declines by 12% each year from an initial 5000. Find the total number of 'animal-years' lived by the population over the first 10 years (i.e. the sum of the population sizes at the start of each of the 10 years), to the nearest whole number.

$a_1 = 5000$, $r = 0.88$. $S_{10} = \frac{5000(1 - 0.88^{10})}{1 - 0.88} = \frac{5000(1 - 0.2785)}{0.12} = \frac{5000(0.7215)}{0.12} \approx 30{,}063$.